ASHRAE Research: Improving the Quality of Life

ASHRAE is the world's foremost technical society in the fields of heating, ventilation, air conditioning, and refrigeration. Its members worldwide are individuals who share ideas, identify needs, support research, and write the industry's standards for testing and practice. The result is that engineers are better able to keep indoor environments safe and productive while protecting and preserving the outdoors for generations to come.

One of the ways that ASHRAE supports its members' and industry's need for information is through ASHRAE Research. Thousands of individuals and companies support ASHRAE Research annually, enabling ASHRAE to report new data about material properties and building physics and to promote the application of innovative technologies.

Chapters in the ASHRAE Handbook are updated through the experience of members of ASHRAE Technical Committees and through results of ASHRAE Research reported at ASHRAE conferences and published in ASHRAE special publications, *ASHRAE Transactions*, and ASHRAE's journal of archival research, *Science and Technology for the Built Environment*.

For information about ASHRAE Research or to become a member, contact ASHRAE, 1791 Tullie Circle N.E., Atlanta, GA 30329; telephone: 404-636-8400; www.ashrae.org.

Preface

The 2017 *ASHRAE Handbook—Fundamentals* covers basic principles and data used in the HVAC&R industry. The ASHRAE Technical Committees that prepare these chapters provide new information, clarify existing content, delete obsolete materials, and reorganize chapters to make the Handbook more understandable and easier to use. An accompanying CD-ROM contains all the volume's chapters in both I-P and SI units.

This edition includes a new chapter:

- Chapter 36, Moisture Management in Buildings, presents data on indoor vapor release and measured indoor/outdoor vapor pressure/concentration differences, and discusses moisture sources and sinks that can reduce materials' durability, as well as the negative effects of insufficient or excessive indoor relative humidity.

Other selected highlights include the following:

- Chapter 7, Fundamentals of Control, has new content on thermostatic valve actuators, placement of sensors, auxiliary control devices, and network architecture.
- Chapter 9, Thermal Comfort, has new content from ASHRAE research project RP-1504 on nonwestern clothing; combined chilled-ceiling, displacement ventilation, and vertical radiant temperature asymmetry effects on sedentary office work; and updates to align with ASHRAE Standard 55-2013.
- Chapter 10, Indoor Environmental Health, has updates on bioaerosols, plus new content on electronic cigarettes and on climate change.
- Chapter 11, Air Contaminants, has new content on particle sizes and settling times, particulate contaminant effects, polymerase chain reaction (PCR) measurement, volatility, mercury, e-cigarettes, and 3D printers.
- Chapter 14, Climatic Design Information, includes new data for 8118 locations worldwide—an increase of 1675 locations from the 2013 edition of the chapter—as a result of ASHRAE research project RP-1699.
- Chapter 15, Fenestration, has updated discussion on U-factor, solar-optical glazing properties, complex glazings and window coverings, tubular daylighting devices (TDDs), and spectrally selective glazing.
- Chapter 16, Ventilation and Infiltration, has been updated and revised for clarity throughout, including recent research results on envelope air leakage.
- Chapter 17, Residential Cooling and Heating Load Calculations, has updates for 2017 climate data and current standards.
- Chapter 18, Nonresidential Cooling and Heating Load Calculations, has new design data for lighting power densities, motors, kitchen equipment, LED lighting, walls and roofs, and an updated example calculation.
- Chapter 19, Energy Estimating and Modeling Methods, extensively revised, has new sections on method development history, using models, uncertainty, thermal loads and model inputs, envelope components, HVAC components, terminal components, low-energy systems, natural and hybrid ventilation, daylighting, passive heating, hybrid inverse method, and model calibration.
- Chapter 20, Space Air Diffusion, has new content on outlet types and characteristics, air curtains, thermal plumes, and air movement in occupied zones.
- Chapter 21, Duct Design, was reorganized for ease of use, and updated for data from the latest version of the *ASHRAE Duct Fitting Database*.
- Chapter 22, Pipe Design, has a new title and now incorporates the content of its sister chapter, Pipes, Tubes, and Fittings, from *HVAC Systems and Equipment*. Also added are content on PEX pipe, plus expanded applications.
- Chapter 24, Airflow Around Buildings, has new content on flow patterns around building groups and isolated buildings, environmental impacts, pollutant dispersion and exhaust reentrainment, pedestrian wind comfort and safety, and wind-driven rain.
- Chapter 30, Thermophysical Properties of Refrigerants, has new or revised data for R-1233zd(E), R-245fa, R-1234ze(E), and R-1234yf.
- Chapter 34, Energy Resources, has extensive updates for new statistics on worldwide energy use and resources.
- Chapter 35, Sustainability, has new content on the water/energy nexus, embodied energy, and climate change.

This volume is published, as a bound print volume and in electronic format on CD-ROM and online, in two editions: one using inch-pound (I-P) units of measurement, the other using the International System of Units (SI).

Corrections to the 2014, 2015, and 2016 Handbook volumes can be found on the ASHRAE website at www.ashrae.org and in the Additions and Corrections section of this volume. Corrections for this volume will be listed in subsequent volumes and on the ASHRAE website.

Reader comments are enthusiastically invited. To suggest improvements for a chapter, **please comment using the form on the ASHRAE website** or, using the cutout page(s) at the end of this volume's index, write to Handbook Editor, ASHRAE, 1791 Tullie Circle, Atlanta, GA 30329, or fax 678-539-2187, or e-mail mowen@ashrae.org.

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Editor

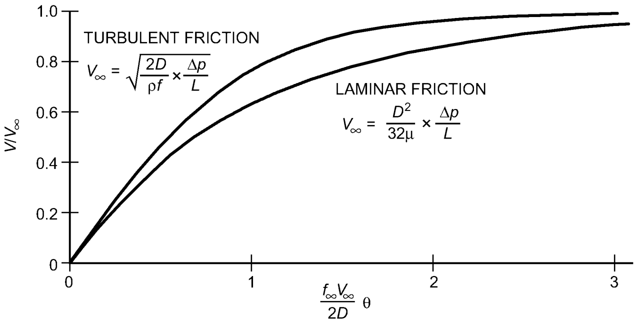

Fig. 20 Temporal Increase in Velocity Following Sudden Application of Pressure

steady-state result for accelerating flows, but appreciably less for decelerating flows. If the friction factor is approximated as constant,

$$\frac{dV}{d\theta} = \frac{\Delta p}{\rho L} - \frac{fV^2}{2D} = A - BV^2 \tag{50}$$

and for the accelerating flow,

$$\theta = \frac{1}{\sqrt{AB}} \tanh^{-1}\left(V\sqrt{\frac{B}{A}}\right) \tag{51}$$

or

$$V = \sqrt{A/B}\ \tanh(\theta\sqrt{AB}) \tag{52}$$

Because the hyperbolic tangent is zero when the independent variable is zero and unity when the variable is infinity, the initial ($V = 0$ at $\theta = 0$) and final conditions are verified. Thus, for long times ($\theta \rightarrow \infty$),

$$V_\infty = \sqrt{A/B} = \sqrt{\frac{\Delta p/\rho L}{f_\infty/2D}} = \sqrt{\frac{\Delta p}{\rho L}\left(\frac{2D}{f_\infty}\right)} \tag{53}$$

which is in accord with Equation (30) when f is constant (the flow regime is the fully rough one of Figure 13). The temporal velocity variation is then

$$V = V_\infty \tanh\,(f_\infty V_\infty \theta/2D) \tag{54}$$

In Figure 20, the turbulent velocity start-up result is compared with the laminar one, where initially the turbulent is steeper but of the same general form, increasing rapidly at the start but reaching V_∞ asymptotically.

Compressibility

All fluids are compressible to some degree; their density depends somewhat on the pressure. Steady liquid flow may ordinarily be treated as incompressible, and incompressible flow analysis is satisfactory for gases and vapors at velocities below about 20 to 40 m/s, except in long conduits.

For liquids in pipelines, a severe pressure surge or water hammer may be produced if flow is suddenly stopped. This pressure surge travels along the pipe at the speed of sound in the liquid, alternately compressing and decompressing the liquid. For steady gas flows in long conduits, the pressure drop along the conduit can reduce gas density enough to increase the velocity. If the conduit is long enough, the velocity may reach the speed of sound, and the Mach number (ratio of flow velocity to the speed of sound) must be considered.

Some compressible flows occur without heat gain or loss (adiabatically). If there is no friction (conversion of flow mechanical energy into internal energy), the process is reversible (isentropic) and follows the relationship

$$p/\rho^k = \text{constant}$$
$$k = c_p/c_v$$

where k, the ratio of specific heats at constant pressure and volume, is 1.4 for air and diatomic gases.

When the elevation term gz is neglected, as it is in most compressible flow analyses, the Bernoulli equation of steady flow, Equation (21), becomes

$$\int \frac{dp}{\rho} + \frac{V^2}{2} = \text{constant} \tag{55}$$

For a frictionless adiabatic process,

$$\int_1^2 \frac{dp}{\rho} = \frac{k}{k-1}\left(\frac{p_2}{\rho_2} - \frac{p_1}{\rho_1}\right) \tag{56}$$

Integrating between upstream station 1 and downstream station 2 gives

$$\frac{p_1}{\rho_1}\left(\frac{k}{k-1}\right)\left[\left(\frac{p_2}{p_1}\right)^{(k-1)/k} - 1\right] + \frac{V_2^2 - V_1^2}{2} = 0 \tag{57}$$

Equation (57) replaces the Bernoulli equation for compressible flows. If station 2 is the stagnation point at the front of a body, $V_2 = 0$, and solving Equation (57) for p_2 gives

$$p_s = p_2 = p_1\left[1 + \left(\frac{k-1}{2}\right)\frac{\rho_1 V_1^2}{kp_1}\right]^{k/(k-1)} \tag{58}$$

where p_s is the stagnation pressure.

Because the speed of sound of the gas is $a = kp/\rho$ and Mach number $M = V/a$, the stagnation pressure in Equation (58) becomes

$$p_s = p_1\left[1 + \left(\frac{k-1}{2}\right)M_1^2\right]^{k/(k-1)} \tag{59}$$

For Mach numbers less than one,

$$p_s = p_1\frac{\rho_1 V_1^2}{2}\left[1 + \frac{M_1}{4} + \left(\frac{2-k}{24}\right)M_1^4 + \cdots\right] \tag{60}$$

When $M = 0$, Equation (60) reduces to the incompressible flow result obtained from Equation (9). When the upstream Mach number exceeds 0.2, the difference is significant. Thus, a pitot tube in air is influenced by compressibility at velocities over about 66 m/s.

Flow Measurement. For isentropic flow through a converging conduit such as a flow nozzle, venturi, or orifice meter, where velocity at the upstream station 1 is small, Equation (57) gives

$$V_2 = \sqrt{\frac{2k}{k-1}\left(\frac{p_1}{\rho_1}\right)\left[1 - \left(\frac{p_2}{p_1}\right)^{(k-1)/k}\right]} \tag{61}$$

The mass flow rate is

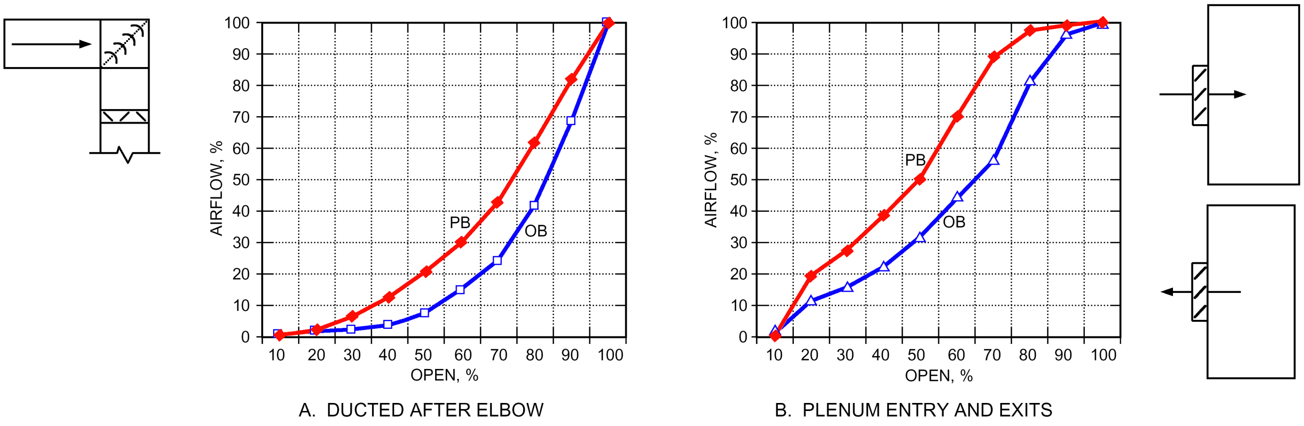

Fig. 15 Inherent Curves for Ducted and Plenum-Mounted Dampers (RP-1157)
Based on data in van Becelaere et al. (2004)

ducted application. These are "inherent" curves, where pressure drop across a damper is held constant as the damper rotates (so it has 100% authority). In real applications, the authority is lower (higher losses of other system components besides the damper). As system pressure losses increase, the curves move up. Note that PB dampers are significantly above linear in most cases.

Figure 15 shows three more applications with PB and OB dampers. The ducted damper has some disturbance and pressure loss ahead of it, to simulate a more realistic situation than those of AMCA 5.3. Nevertheless, the response curves are similar to AMCA 5.3. The plenum entry dampers show irregular results. Again, these are inherent curves, and lower authority causes the curves to move up toward more flow at smaller angles.

The curves shown here are typical, but do not represent every scenario. Thousands of installation variations exist, and slight variations in response always occur. For additional application examples and greater detail, see ASHRAE research project RP-1157 (van Becelaere et al. 2004).

Application. Dampers require engineering to achieve defined goals. A common application is a flow control damper, which modulates airflow. The curves in Figure 13 can be used to pick a damper with a pressure drop and authority that provides near-linear response. Another common application is economizer outdoor air, return air, and exhaust air dampers. Selection of these dampers depends on the system design, as discussed in ASHRAE *Guideline* 16-2014.

Damper leakage is a concern, particularly where tight shutoff is necessary to significantly reduce energy consumption. Also, outdoor air dampers in cold climates must close tightly to prevent coils and pipes from freezing. Low-leakage dampers cost more and require larger actuators because of friction of the seals in the closed position; however, the energy savings offset the extra cost.

Actuators. Either electricity or compressed air is used to actuate dampers.

Pneumatic damper actuators are similar to pneumatic valve actuators, except that they have a longer stroke or the stroke is increased by a multiplying lever. Increasing air pressure produces a linear shaft motion, which, through a linkage, moves the crank arm to open or close the dampers. Releasing air pressure allows a spring to return the actuator. Double-acting actuators without springs are available also.

Electric damper actuators can be proportional or two-position. They can be spring return or nonspring. The simplest form of control is a floating three-point controller, in which contact closures drive the motor clockwise or counterclockwise. In addition, a variety of standard electronic signals from electronic controllers or DDC systems, such as 4 to 20 mA, 2 to 10 V (dc), or 0 to 10 V (dc), can be used to control proportionally actuated dampers.

Most modern actuators are electronic and use more sophisticated methods of control and operation. They are inherently positive positioning and may have communications capabilities similar to industrial-process-quality actuators.

A two-position spring-return actuator moves in one direction when power is applied to its internal windings. When no power is present, the actuator returns (via spring force) to its fail-safe (normal) position. Depending on how the actuator is connected, this action opens or closes the dampers. A proportional actuator may also have spring-return action.

Mounting. Damper actuators are mounted in different ways, depending on the size and accessibility of the damper, and the power required to move the damper. The most common method of mounting electric actuators is directly over the damper shaft (direct coupling) with no external linkage. Actuators can also be mounted in the airflow on the damper frame and be linked directly to a damper blade (though this may deform blades and make access for repairs difficult), or mounted outside the duct and connected to a crank arm attached to a shaft extension of one of the blades. Mounting methods that link the actuator to a single blade are not recommended because of the potential to bend the single drive blade or to have the blade-to-blade linkages get out of adjustment.

Large dampers may require two or more actuators, which are usually mounted at separate points on the damper. An alternative is to install the damper in two or more sections, each section being controlled by a single damper actuator. Positive positioners may be required for proper sequencing, as with a small damper controlled to maintain healthy indoor air quality, with a large damper being independently controlled for economy-cycle free cooling.

Pneumatic Positive (Pilot) Positioners

Where accurate positioning of a modulating pneumatic damper or valve is required, use positive positioners. A positive positioner provides up to full supply control air pressure to the actuator for any change in position required by the controller. A pneumatic actuator-without a positioner may not respond quickly or accurately enough to small changes in control pressure caused by friction in the actuator or load, or to changing load conditions such as wind acting on a damper blade. A positive positioner provides finite and repeatable positioning change and allows adjustment of the control range (spring range of the actuator) to provide a proper sequencing control of two or more control devices.